

Quantum Logic and Gates

Logic for Multiple Qubits

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No-Cloning Theorem

- Can we perform a FANOUT in a quantum computer?

- Try to copy a qubit in an unknown state

$$|\psi\rangle = (\alpha|0\rangle + \beta|1\rangle)$$

- Start with a scratch ancilla qubit to store the output.

$$(\alpha|0\rangle + \beta|1\rangle)|0\rangle = \alpha|00\rangle + \beta|10\rangle$$

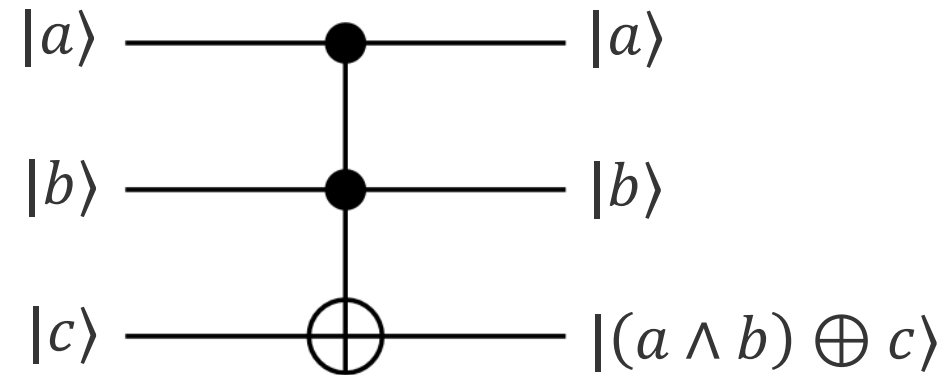
$$CX(\alpha|00\rangle + \beta|10\rangle) = \alpha|00\rangle + \beta|11\rangle$$

- But this $\neq |\psi\rangle|\psi\rangle$!

Quantum logic gates

- How do we make AND and OR gates reversibly?
- We use a Toffoli gate.
- Toffoli (**CCX**) negates the third qubit iff the first two qubits are both $|1\rangle$:
 $|a, b, c\rangle \rightarrow |a, b, (a \wedge b) \oplus c\rangle$

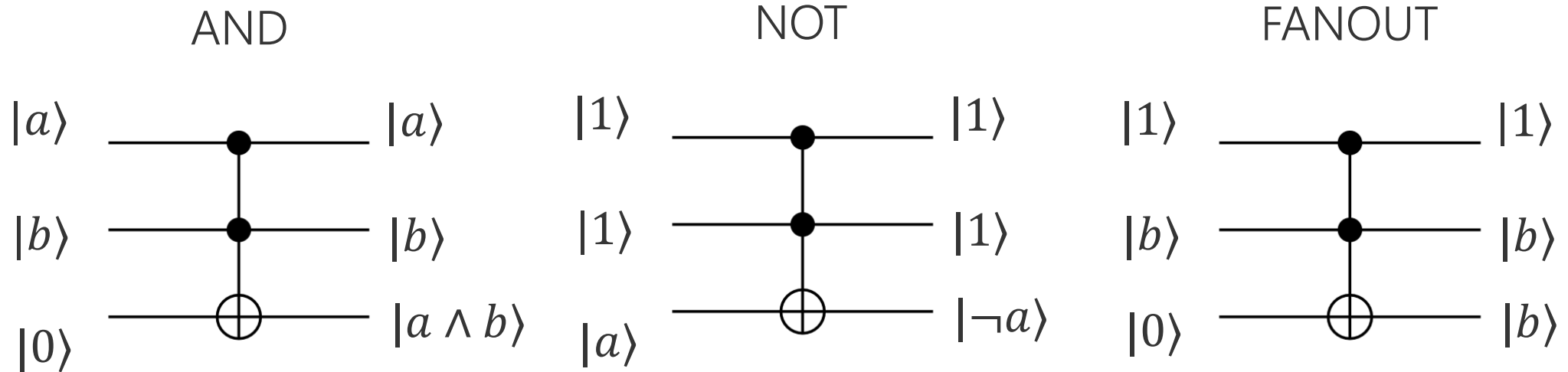
XOR



$$\begin{pmatrix} 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \end{pmatrix}$$

Universality of Toffoli

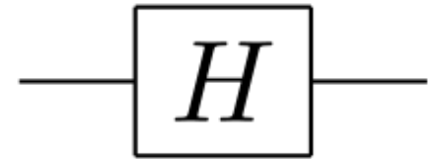
- A Toffoli gate can be used to simulate AND, NOT, and FANOUT gates



Preparing a state in superposition

- We need to be able to prepare a state in, say, $|+\rangle$
- The Hadamard gate, H , is the trick:

$$H = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}.$$



$$\begin{pmatrix} 1 \\ 0 \end{pmatrix} \rightarrow \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \end{pmatrix} \quad \begin{pmatrix} 0 \\ 1 \end{pmatrix} \rightarrow \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ -1 \end{pmatrix}$$

$$\begin{pmatrix} c_0 \\ c_1 \end{pmatrix} \rightarrow \frac{1}{\sqrt{2}} \begin{pmatrix} c_0 + c_1 \\ c_0 - c_1 \end{pmatrix}$$

$$|0\rangle \rightarrow \frac{1}{\sqrt{2}} (|0\rangle + |1\rangle), |1\rangle \rightarrow \frac{1}{\sqrt{2}} (|0\rangle - |1\rangle)$$

Changing the phase

- How do we modify phase?

$$Z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$$
$$Z(\alpha|0\rangle + \beta|1\rangle) = \alpha|0\rangle - \beta|1\rangle$$

$$S = \begin{pmatrix} 1 & 0 \\ 0 & i \end{pmatrix}$$
$$S(\alpha|0\rangle + \beta|1\rangle) = \alpha|0\rangle + i\beta|1\rangle$$

$$Y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}$$
$$Y(\alpha|0\rangle + \beta|1\rangle) = -i\beta|0\rangle + i\alpha|1\rangle$$

$$I = -iXYZ$$

$$XZ = -ZX$$

General single-qubit unitary gate

- A single-qubit gate can in general be written as

$$U = e^{i\gamma}V$$

$$V = \begin{pmatrix} \alpha & -\beta^* \\ \beta & \alpha^* \end{pmatrix}$$

Satisfies normalization condition: $|\alpha|^2 + |\beta|^2 = 1$

An important rotation gate

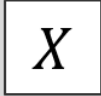
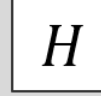
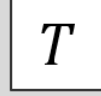
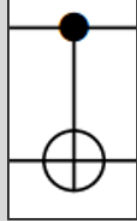
$$T = \begin{pmatrix} 1 & 0 \\ 0 & e^{i\pi/4} \end{pmatrix}$$

We can view S and Z also as rotation gates.

$$e^{i\pi} = -1, \quad e^{\frac{i\pi}{2}} = i$$
$$Z = \begin{pmatrix} 1 & 0 \\ 0 & e^{i\pi} \end{pmatrix}$$

$$T^2 = S, \quad S^2 = Z, \quad T^4 = Z$$

Quantum gates

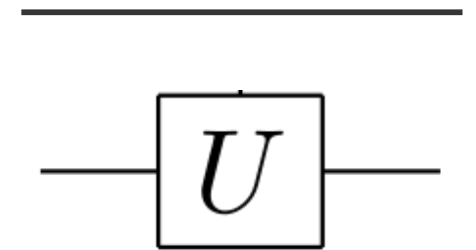
Gate	Matrix	Symbol
Pauli X	$\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$	
Pauli Z	$\begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$	
Pauli Y	$\begin{bmatrix} 0 & -i \\ i & 0 \end{bmatrix}$	
Phase S	$\begin{bmatrix} 1 & 0 \\ 0 & i \end{bmatrix}$	
Hadamard H	$\frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}$	
T = " $\pi/8$ "	$\begin{bmatrix} 1 & 0 \\ 0 & e^{i\pi/4} \end{bmatrix}$	
Controlled Not CNOT	$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{bmatrix}$	

Example:

$$U = \begin{pmatrix} u_0 & u_1 \\ u_2 & u_3 \end{pmatrix}$$

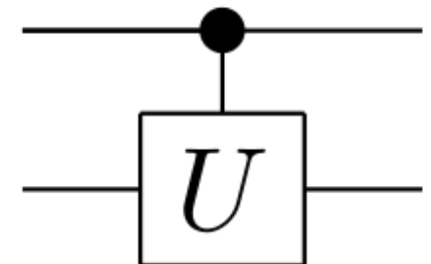
Apply U to one of two qubits in a system. I is applied to the other qubit.

$$I \otimes U = \begin{pmatrix} u_0 & u_1 & 0 & 0 \\ u_2 & u_3 & 0 & 0 \\ 0 & 0 & u_0 & u_1 \\ 0 & 0 & u_2 & u_3 \end{pmatrix}$$



This is different than controlling the application of U based on a qubit:

$$CU = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & u_0 & u_1 \\ 0 & 0 & u_2 & u_3 \end{pmatrix}$$



Summary

- Quantum gates: Quantum operations and state evolution must be reversible; the only condition on a quantum gate is that it must be unitary.
- No-cloning theorem: An arbitrary quantum state cannot be copied.
- Universal Set of Gates: CNOT, H and T